

BAIRNSDALE, Australia

WMO: 949120

Lat: 37.8817S

Lon: 147.5669E

Elev: 50

StdP: 100.73

Time Zone: 10.00 (AUS)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	2.4	-0.5	3.6	6.5	0.4	3.9	6.8	12.3	13.2	11.1	13.3	2.5	310	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	33.2	19.4	30.2	19.2	27.7	18.7	21.3	28.4	20.5	26.7	19.7	25.2	6.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	22.2	18.8	13.7	21.9	17.8	12.9	21.5	62.2	28.9	59.2	26.6	56.5	25.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.1	8.9	DB	-1.3	39.9	1.2	2.7	-2.2	41.9	-2.9	43.4	-3.6	44.9	-4.5	46.9	
			WB	-1.5	23.2	1.3	1.1	-2.5	24.0	-3.2	24.6	-3.9	25.2	-4.8	26.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	19.3	19.0	17.6	14.8	12.1	9.8	9.5	10.1	11.7	13.5	15.7	17.3
	DBStd	4.45	3.64	3.08	3.10	2.71	2.30	2.06	1.99	2.14	2.62	3.02	3.34	3.37
	HDD10.0	101	0	0	0	1	4	28	33	24	8	2	0	0
	HDD18.3	1708	29	25	53	112	193	256	274	256	200	156	96	61
	CDD10.0	1620	288	251	234	144	71	22	18	27	59	109	171	227
	CDD18.3	186	58	43	29	5	1	0	0	0	1	4	16	29
	CDH23.3	1990	607	361	276	64	4	0	0	1	20	95	216	346
	CDH26.7	784	281	133	101	15	0	0	0	0	3	27	79	144
(14) Wind	WSAvg	4.1	4.2	4.1	4.0	3.6	3.8	3.7	4.0	4.2	4.4	4.3	4.3	4.4
(15) Precipitation	PrecAvg	675	55	57	46	60	40	70	45	43	58	63	74	64
	PrecMax	869	127	154	118	189	107	294	108	71	165	122	161	137
	PrecMin	448	14	4	12	10	15	9	12	5	25	16	22	24
	PrecStd	109	29	35	24	49	20	70	28	17	31	29	33	29
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	35.2	33.7	28.8	23.3	19.5	19.1	21.0	26.0	30.0	33.1	35.7
		MCWB	20.9	20.4	19.5	17.8	15.1	13.2	11.8	12.7	15.1	17.5	18.3	19.8
	2%	DB	33.3	30.5	29.5	24.9	20.7	17.1	17.0	18.5	22.4	25.9	28.7	30.6
		MCWB	20.0	19.7	18.6	16.7	13.9	11.6	10.8	11.6	14.0	15.9	18.2	19.2
	5%	DB	29.5	27.6	26.2	22.3	18.6	15.5	15.4	16.8	19.8	23.0	25.5	27.1
		MCWB	19.7	19.6	18.0	15.7	13.2	10.9	10.1	10.8	13.1	15.0	17.3	18.5
	10%	DB	26.1	25.0	23.6	20.1	16.9	14.2	14.0	15.2	17.6	20.0	22.4	24.0
		MCWB	19.0	19.0	17.4	15.0	12.4	10.4	9.5	10.2	12.0	14.1	16.4	17.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	22.2	21.2	18.5	16.1	14.0	12.7	13.6	16.3	18.7	20.5	21.5
		MCDB	32.3	29.7	26.6	25.4	20.7	17.1	16.1	19.3	23.1	25.3	27.9	29.7
	2%	WB	21.4	21.2	20.1	17.4	14.7	12.8	11.6	12.3	14.7	16.8	19.1	20.1
		MCDB	29.2	27.7	25.0	22.3	19.1	15.7	14.9	17.1	20.8	23.0	25.6	27.2
	5%	WB	20.5	20.3	19.1	16.6	13.7	11.8	10.8	11.3	13.5	15.6	17.9	19.1
		MCDB	27.2	25.2	24.3	20.6	17.4	14.5	14.5	15.4	18.7	21.1	23.7	25.5
	10%	WB	19.5	19.4	18.2	15.7	12.9	10.9	10.1	10.6	12.5	14.5	16.9	18.1
		MCDB	25.3	23.6	23.1	19.3	16.5	13.5	13.4	14.4	16.8	19.6	21.8	23.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	10.7	11.2	10.7	9.7	9.1	9.3	10.0	11.0	11.5	11.0	11.4
		MCDBR	17.9	15.9	16.5	14.8	12.2	10.9	11.4	12.6	15.3	16.9	17.2	17.7
		MCWBR	6.5	6.5	7.1	7.2	6.8	6.6	6.6	7.1	8.0	8.1	7.6	7.4
	5% WB	MCDBR	14.6	12.2	12.8	12.1	10.9	9.9	9.9	11.4	13.7	14.8	14.4	14.4
		MCWBR	6.0	5.6	6.1	6.4	6.5	6.6	6.2	7.0	7.7	7.7	7.1	6.8
(40) Clear-Sky Solar Irradiance	taub		0.397	0.368	0.340	0.330	0.307	0.295	0.294	0.304	0.326	0.340	0.350	0.383
	taud		2.378	2.491	2.560	2.562	2.602	2.627	2.619	2.584	2.522	2.506	2.510	2.411
	Ebn at Noon		936	940	928	876	840	824	847	892	928	958	976	954
	Edh at Noon		127	109	94	82	68	62	66	79	96	106	111	125
(44) All-Sky Solar Radiation	RadAvg		6.74	5.81	4.82	3.47	2.45	1.94	2.23	3.08	4.26	5.46	6.21	6.90
	RadStd		0.44	0.40	0.34	0.24	0.12	0.13	0.12	0.14	0.31	0.35	0.43	0.42

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.46	N/A	N/A	+1.44	N/A	N/A	N/A	-103	+169	+73
(47)	Regional (27 neighbors)	+0.43	N/A	N/A	N/A	+0.43	N/A	N/A	-102	+147	+55

Nomenclature: See separate page